TO THE TRAINER

This PowerPoint presentation can be used to train people about the basics of compressors and compressed air systems. The information on the slides is the minimum information that should be explained. The trainer notes for each slide provide more detailed information, but it is up to the trainer to decide if and how much of this information is presented also.

Additional materials that can be used for the training session are available on **www.energyefficiencyasia.org** under “Energy Equipment” and include:

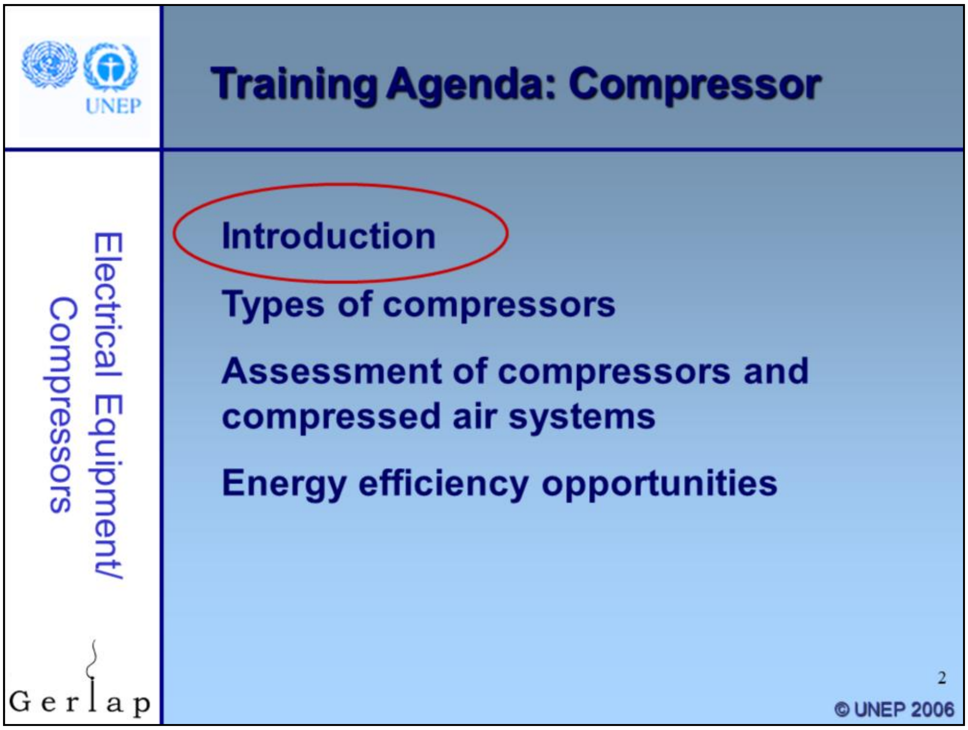
•**Textbook chapter** on this energy equipment that forms the basis of this PowerPoint presentation but has more detailed information

•**Quiz** – ten multiple choice questions that trainees can answer after the training session

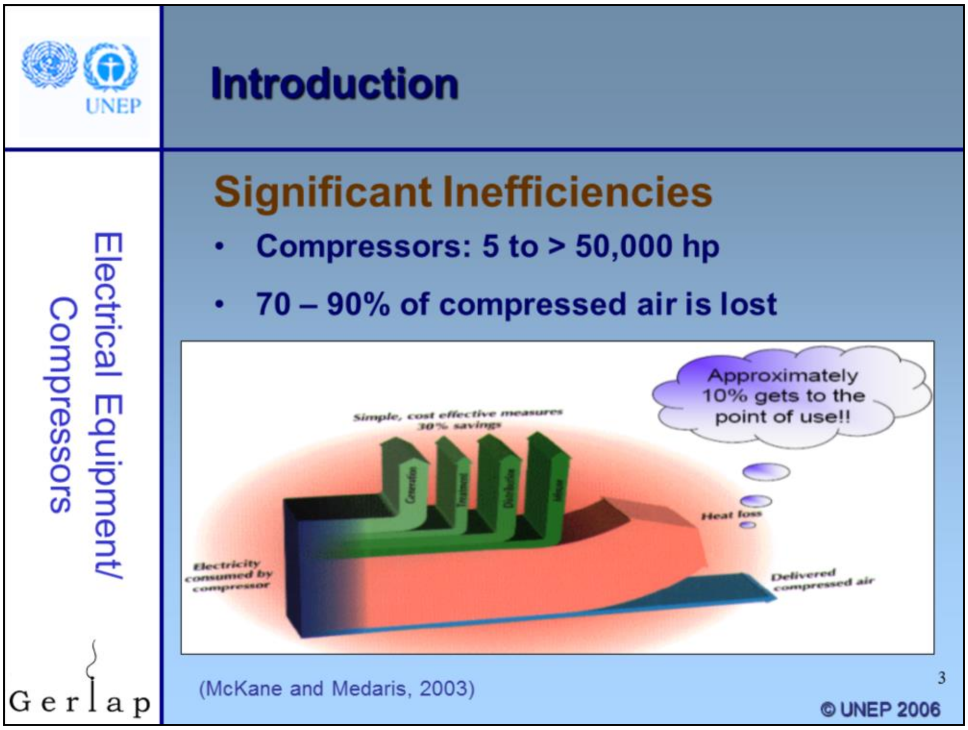
•**Workshop exercise** – a practical calculation related to this equipment

•**Option checklist** – a list of the most important options to improve energy efficiency of this equipment

•**Company case studies** – participants of past courses have given the feedback that they would like to hear about options implemented at companies for each energy equipment. More than 200 examples are available from 44 companies in the cement, steel, chemicals, ceramics and pulp & paper sectors



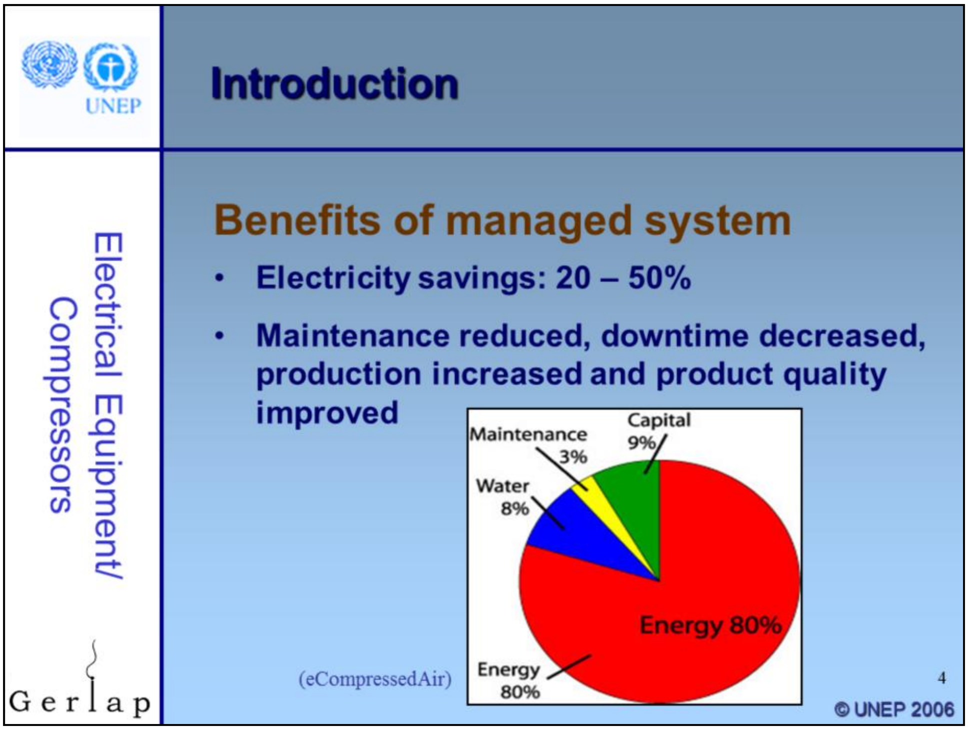
This first section covers an introduction to compressors and compressed air systems.



•Industrial plants use compressed air throughout their production operations, which is produced by compressed air units ranging from 5 horsepower (hp) to over 50,000 hp.

•The US Department of Energy reports that 70 to 90 percent of compressed air is lost in the form of unusable heat, friction, misuse and noise.

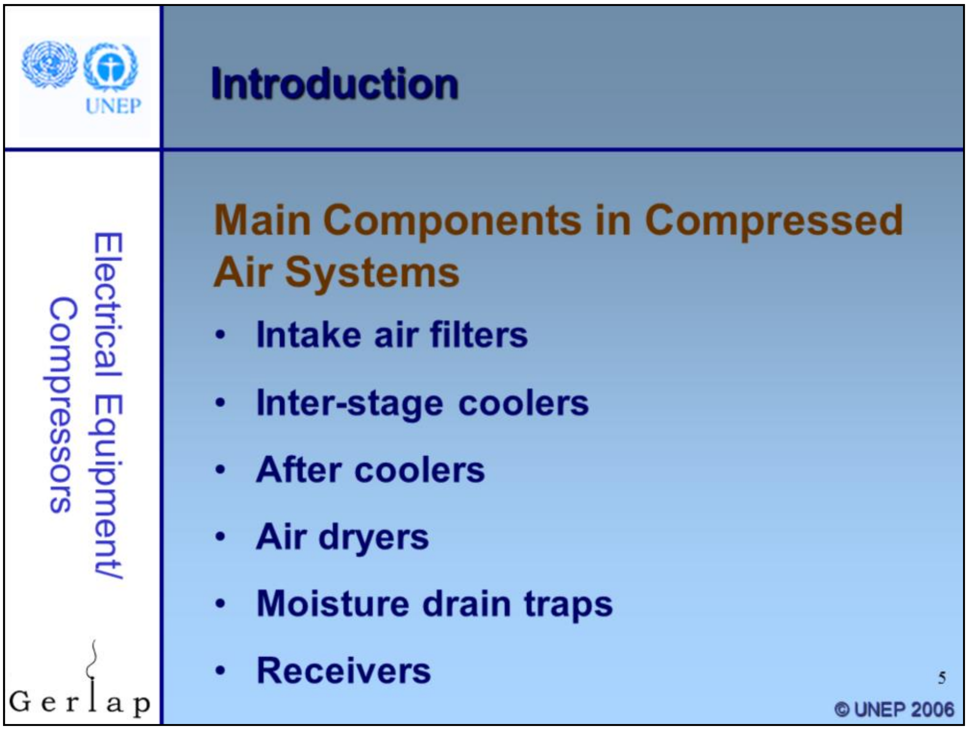
•For this reason, compressors and compressed air systems are important areas to improve energy efficiency at industrial plants.



•It is worth noting that the running cost of a compressed air system is far higher than the cost of a compressor itself.

•Energy savings from system improvements can range from 20 to 50 percent or more of electricity consumption, resulting in thousands to hundreds of thousands of dollars.

•A properly managed compressed air system can save energy, reduce maintenance, decrease downtime, increase production throughput, and improve product quality



Compressed air systems consist of following major components:

•Intake Air Filters that prevent dust from entering a compressor. Dust causes sticking valves, scoured cylinders, excessive wear etc.

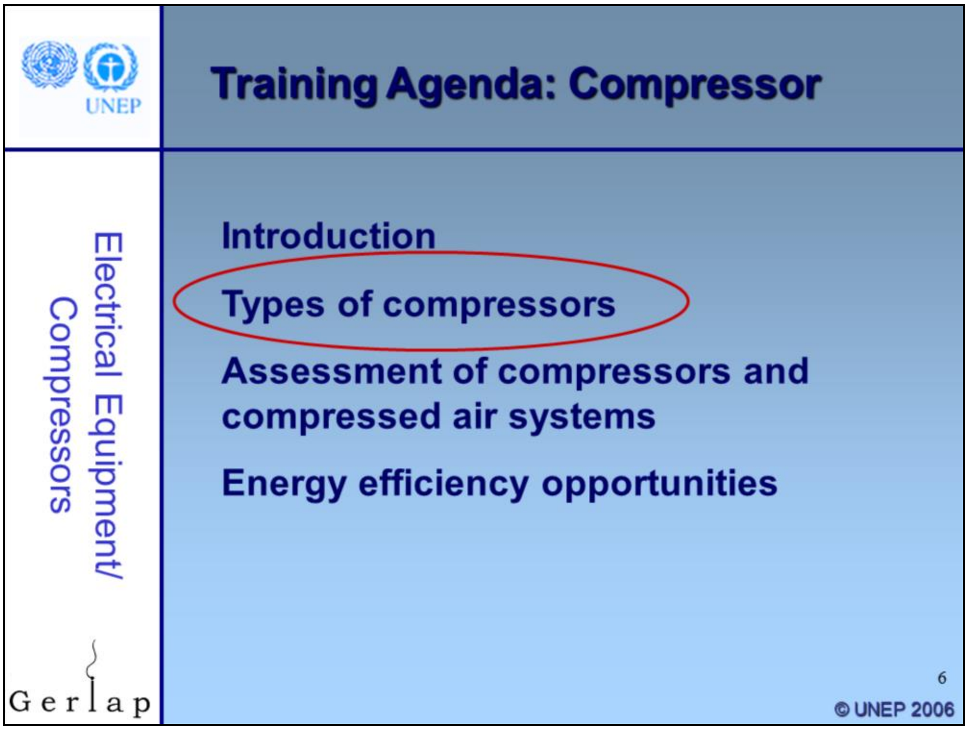
•Inter-stage Coolers that reduce the temperature of the air before it enters the next stage to reduce the work of compression and increase efficiency. They are normally water-cooled.

•After-Coolers with the objective is to remove the moisture in the air by reducing the temperature in a water-cooled heat exchanger.

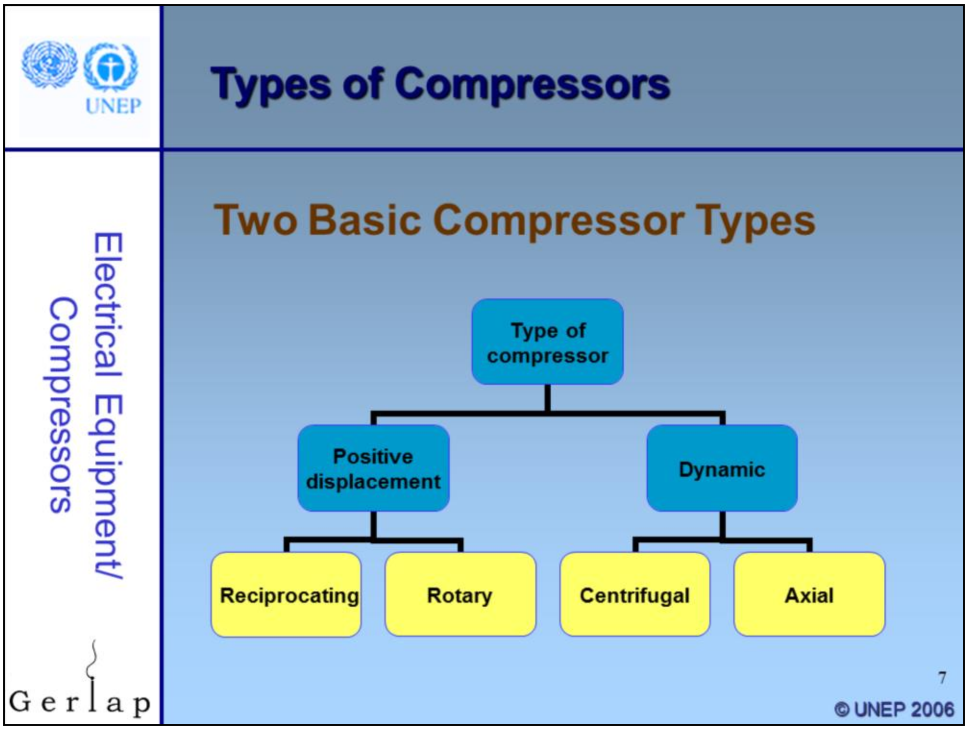
•Air-dryers that remove the remaining traces of moisture after after-cooler as equipment has to be relatively free of any moisture.

•Moisture drain traps that are used for removal of moisture in the compressed air. These traps resemble steam traps. Various types of traps used are manual drain cocks, timer based / automatic drain valves etc.

•Receivers that are provided as storage and smoothening pulsating air output - reducing pressure variations from the compressor



Types of compressors.

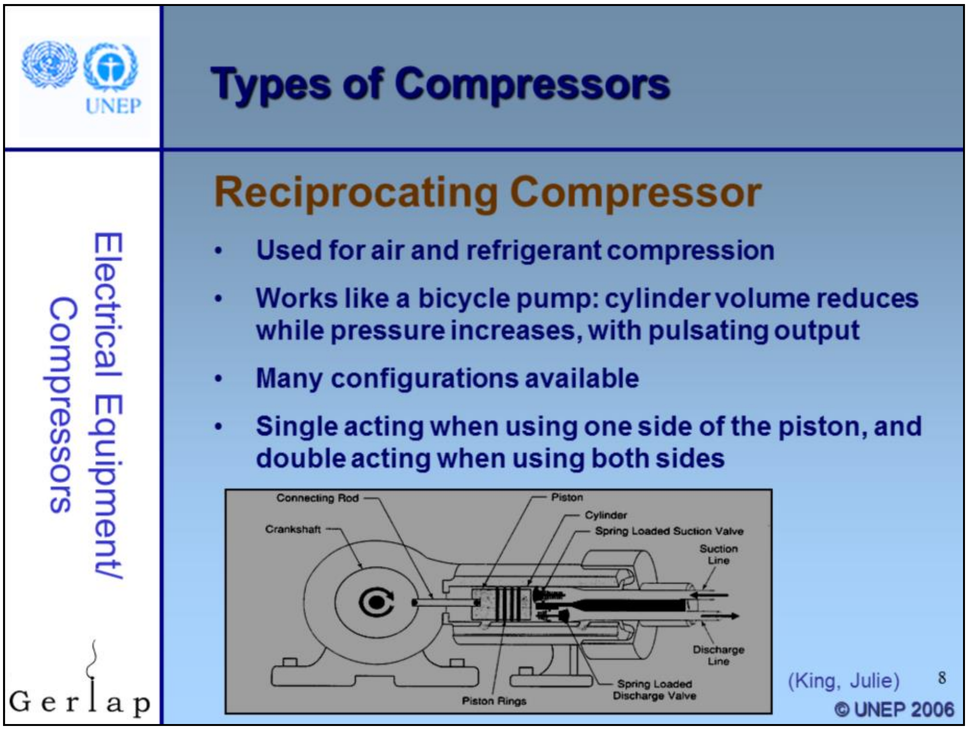


There are two basic compressor types: positive-displacement and dynamic.

•In the positive-displacement type, a given quantity of air or gas is trapped in a compression chamber and the volume it occupies is mechanically reduced, causing a corresponding rise in pressure prior to discharge. At constant speed, the air flow remains essentially constant with variations in discharge pressure.

•Dynamic compressors impart velocity energy to continuously flowing air or gas by means of impellers rotating at very high speeds. The velocity energy is changed into pressure energy both by the impellers and the discharge volutes or diffusers. In the centrifugal-type dynamic compressors, the shape of the impeller blades determines the relationship between air flow and the pressure (or head) generate

We will now go through the various types of compressors.



•In industry, reciprocating compressors are the most widely used type for both air and refrigerant compression.

•They work on the principles of a bicycle pump and are characterized by a flow output that remains nearly constant over a range of discharge pressures. Also, the compressor capacity is directly proportional to the speed. The output, however, is a pulsating one.

•Reciprocating compressors are available in many configurations

- •The four most widely used are horizontal, vertical, horizontal balance-opposed and tandem
- •air-cooled or water-cooled
- •lubricated and non-lubricated
- •may be packaged
- •provide a wide range of pressure and capacity selections.

•The reciprocating air compressor is considered single acting when the compressing is accomplished using only one side of the piston. A compressor using both sides of the piston is considered double acting.

UNEP

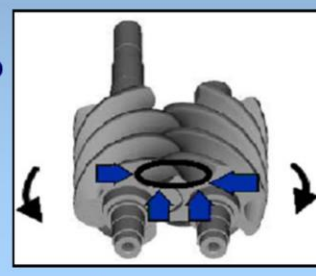
Electrical Equipment/ Compressors

Gerlap

Types of Compressors

Rotary Compressor

- Rotors instead of pistons: continuous discharge
- Benefits: low cost, compact, low weight, easy to maintain
- Sizes between 30 – 200 hp
- Types
    - Lobe compressor
    - Screw compressor
    - Rotary vane / Slide vane



Screw compressor

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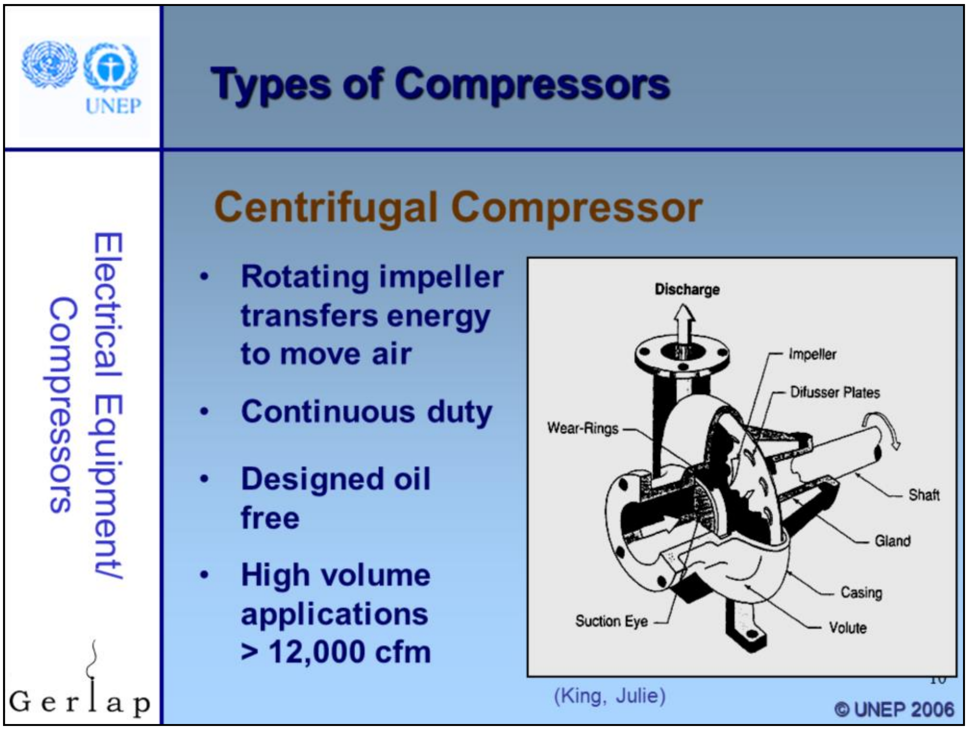
•Rotary compressors have rotors in place of pistons and give a continuous pulsation free discharge. They operate at high speed and generally provide higher throughput than reciprocating compressors. Their capital costs are low, they are compact in size, have low weight, and are easy to maintain. For this reason they have gained popularity with industry. They are most commonly used in sizes from about 30 to 200 hp or 22 to 150 kW.

•Types of rotary compressors include: Lobe compressor, Screw compressor, Rotary vane / sliding-vane. The picture shows a screw compressor

•Rotary screw compressors may be air or water-cooled. Since the cooling takes place right inside the compressor, the working parts never experience extreme operating temperatures. The rotary compressor, therefore, is a continuous duty, air cooled or water cooled compressor package.



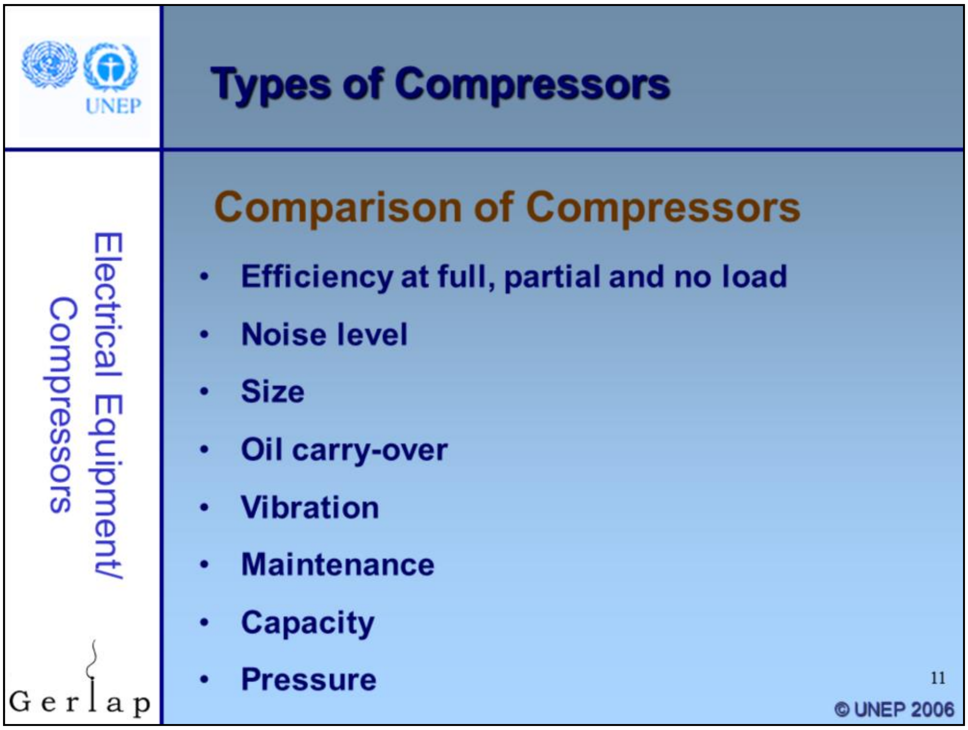
•The centrifugal air compressor is a dynamic compressor, which depends on transfer of energy from a rotating impeller to the air. The rotor accomplishes this by changing the momentum and pressure of the air. This momentum is converted to useful pressure by slowing the air down in a stationary diffuser.

•The centrifugal air compressor is an oil free compressor by design. The oil lubricated running gear is separated from the air by shaft seals and atmospheric vents.

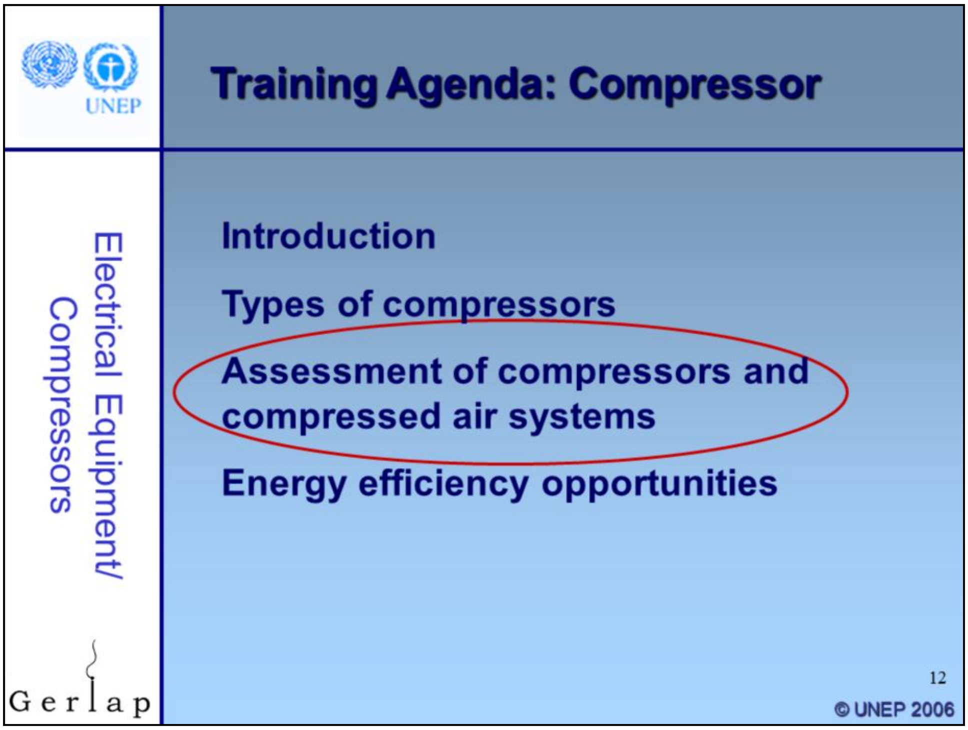
•The centrifugal is a continuous duty compressor, with few moving parts, that is particularly suited to high volume applications-especially where oil free air is required.

•Centrifugal air compressors are water-cooled and may be packaged; typically the package includes the after-cooler and all controls.

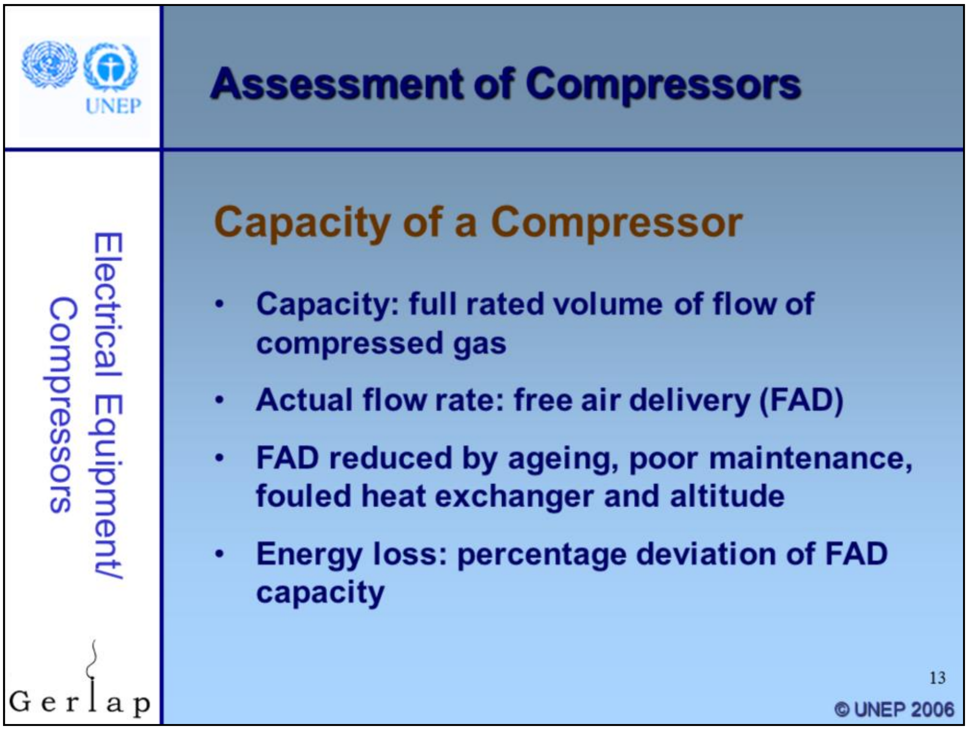
•These compressors have appreciably different characteristics as compared to reciprocating machines. A small change in compression ratio produces a marked change in compressor output and efficiency. Centrifugal machines are better suited for applications requiring very high capacities, typically above 12,000 cfm (cubic feet per minute).



•These factors are important when selecting a compressor. The chapter includes a table comparing different compressors on these factors.



•Assessment of compressors and compressed air systems.

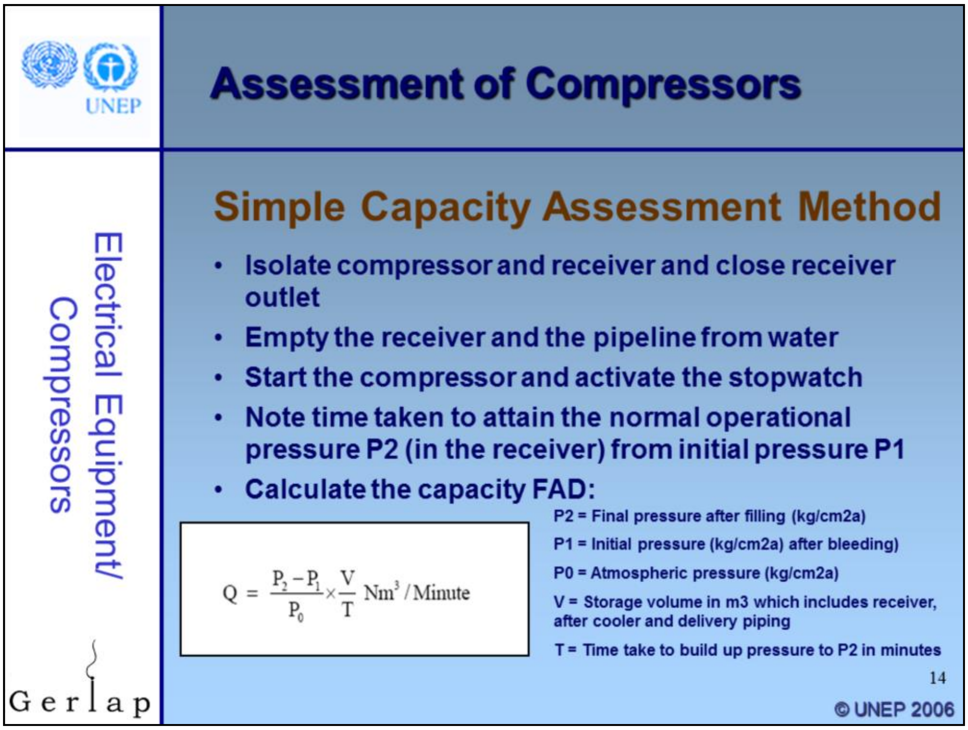


•The capacity of a compressor is the full rated volume of flow of gas compressed and delivered under conditions of total temperature, total pressure, and composition prevailing at the compressor inlet.

•It sometimes means actual flow rate, rather than rated volume of flow. This is also called free air delivery (FAD) i.e. air at atmospheric conditions at any specific location. This term does not mean air delivered under identical or standard conditions because the altitude, barometer, and temperature may vary at different localities and at different times.

•Due to ageing of the compressors and inherent inefficiencies in the internal components, the free air delivered may be less than the design value, despite good maintenance practices. Sometimes, other factors such as poor maintenance, fouled heat exchanger and effects of altitude also tend to reduce free air delivery. In order to meet the air demand, the inefficient compressor may have to run for more time, thus consuming more power than actually required.

•The power wastage depends on the percentage deviation of FAD capacity. For example, a worn out compressor valve can reduce the compressor capacity by as much as 20 percent. A periodic assessment of the FAD capacity of each compressor has to be carried out to check its actual capacity. If the deviations are more than 10 percent, corrective measures should be taken to rectify the same.



We will go through how to perform a simple capacity assessment in a shop floor:

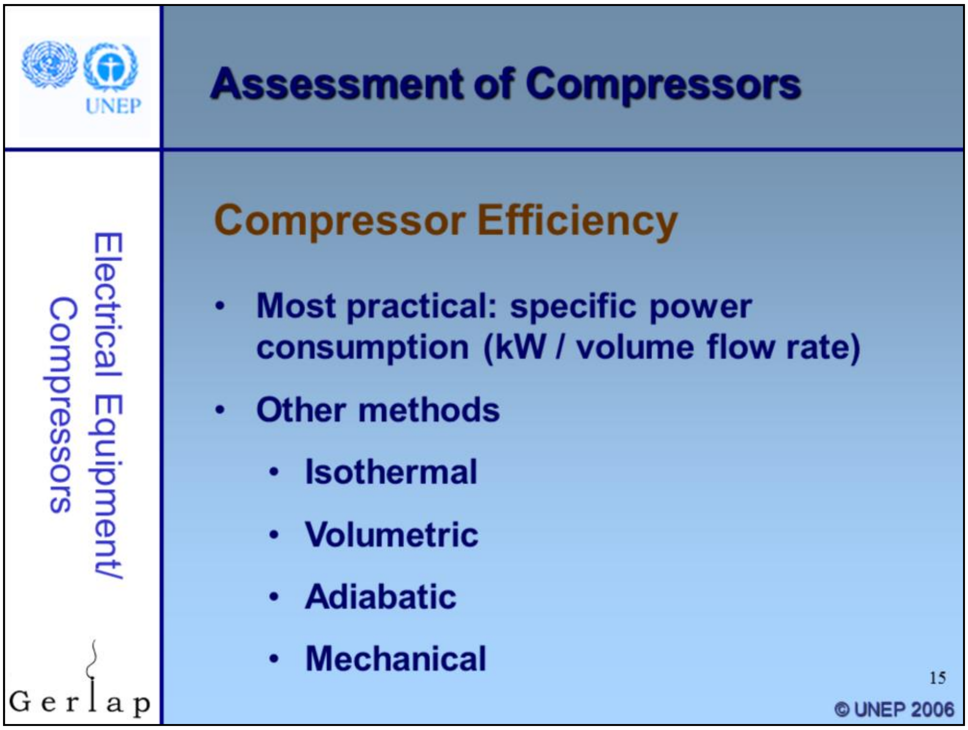
•Isolate the compressor along with its individual receiver that are to be taken for a test from the main compressed air system by tightly closing the isolation valve or blanking it, thus closing the receiver outlet.

•Open the water drain valve and drain out water fully and empty the receiver and the pipeline. Make sure that the water trap line is tightly closed once again to start the test.

•Start the compressor and activate the stopwatch.

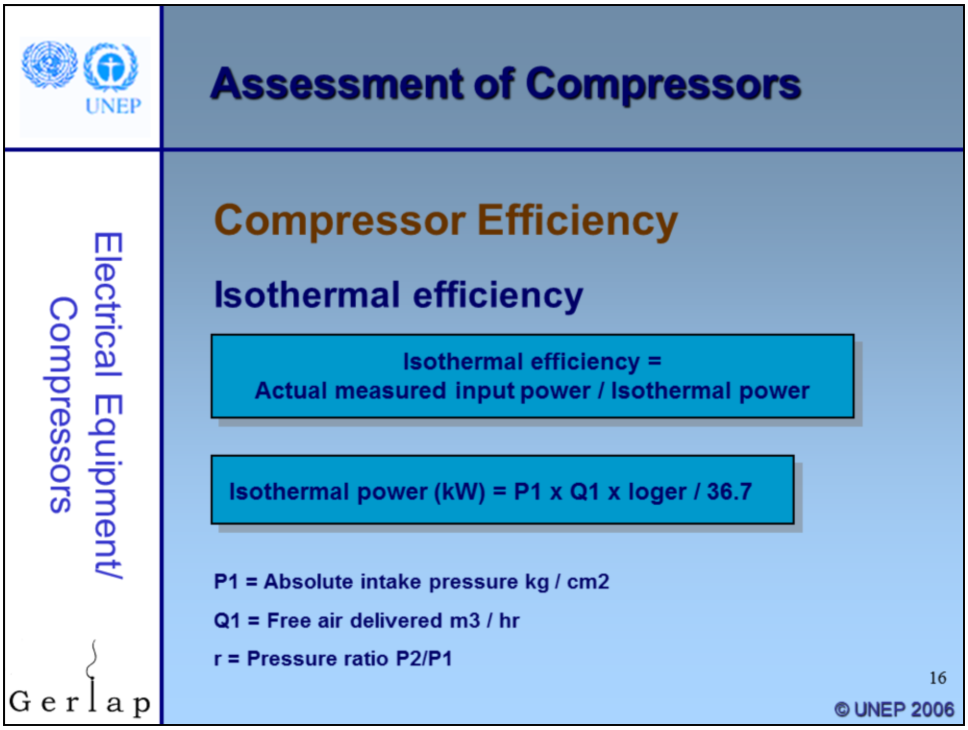
•Note the time taken to attain the normal operational pressure P2 (in the receiver) from initial pressure P1.

•Calculate the capacity as per the formulae given. FAD is to be corrected by a factor (273 + t1) / (273 + t2)



•For practical purposes, the most effective guide in comparing compressor efficiencies is the specific power consumption, i.e. kW/volume flow rate, for different compressors that would provide identical duty.

•There are several different measures of compressor efficiency that are commonly used including volumetric efficiency, adiabatic efficiency, isothermal efficiency and mechanical efficiency. We will only discuss isothermal and volumetric efficiency calculation methods here

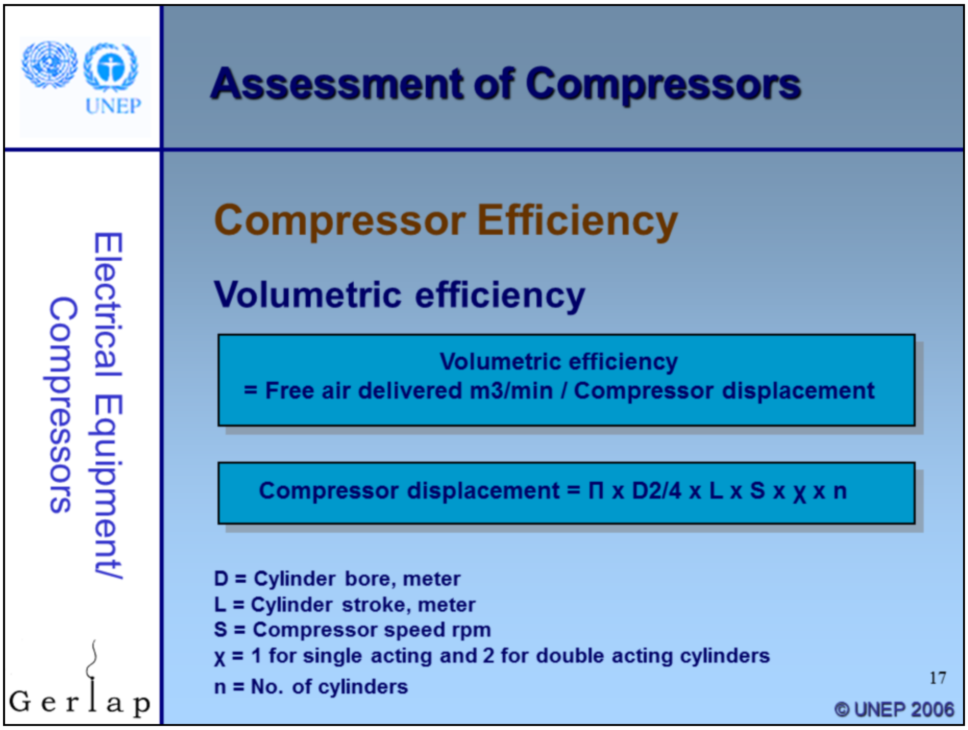


•The reported value of efficiency is normally the isothermal efficiency. This is an important consideration when selecting compressors based on reported values of efficiency.

•Isothermal efficiency is calculated as follows:

- •Isothermal Efficiency=Actual measured input power / Isothermal Power
- •Isothermal power (kW) = P1 x Q1 x loger/36.7
- •Where P1 = Absolute intake pressure kg/ cm2; Q1 = Free air delivered m3/hr; and r = Pressure ratio P2/P1.

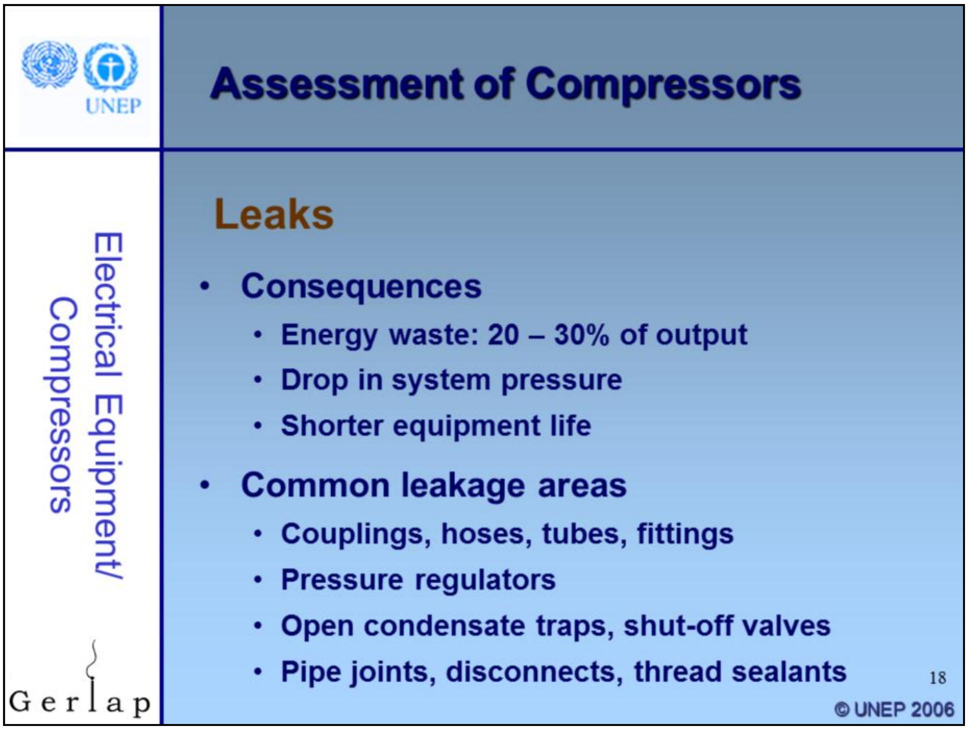
•The calculation of isothermal power does not include power needed to overcome friction and generally gives an efficiency that is lower than adiabatic efficiency.



•Volumetric efficiency = Free air delivered m3/min / Compressor displacement

•Compressor Displacement = Π x D2/4 x L x S x χ x n

•Where D = Cylinder bore, meter; L = Cylinder stroke, meter; S = Compressor speed rpm; χ = 1 for single acting and 2 for double acting cylinders; and n = No. of cylinders



•A system of distribution pipes and regulators convey compressed air from the central compressor plant to process areas. This system includes various isolation valves, fluid traps, intermediate storage vessels, and even heat trace on pipes to prevent condensation or freezing in lines exposed to the outdoors. Pressure losses in distribution typically are compensated for by higher pressure at the compressor discharge.

•At the intended point of use, a feeder pipe with a final isolation valve, filter, and regulator carries the compressed air to hoses that supply processes or pneumatic tools.

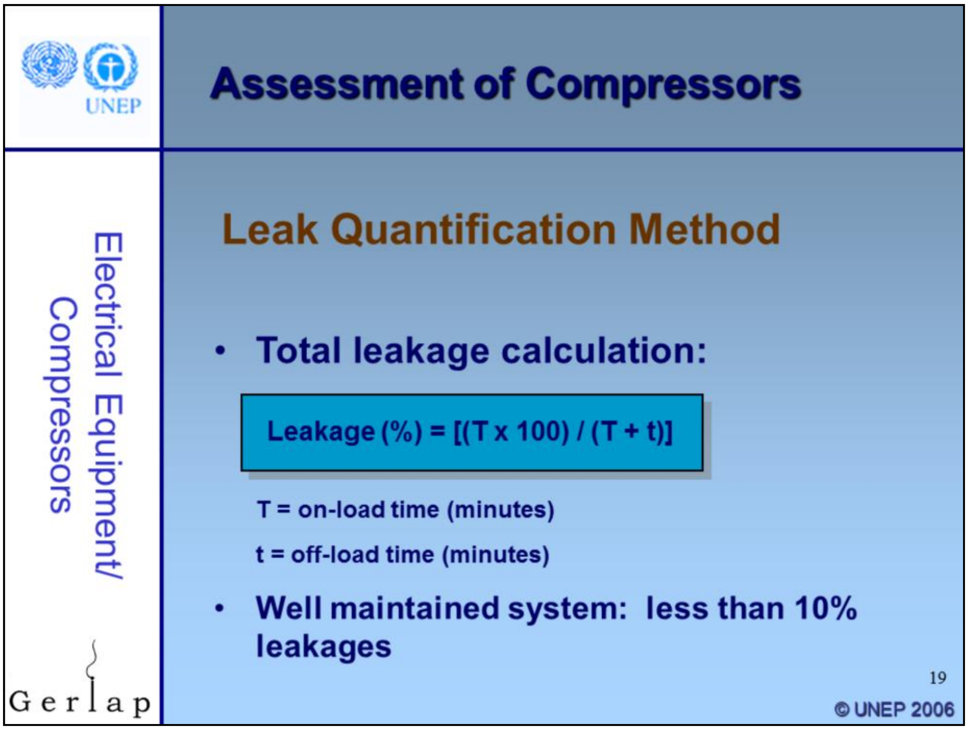
•Leaks can be a significant source of wasted energy in an industrial compressed air system, sometimes wasting 20 to 30 percent of a compressor's output. A typical plant that has not been well maintained will likely have a leak rate equal to 20 percent of total compressed air production capacity. On the other hand, proactive leak detection and repair can reduce leaks to less than 10 percent of compressor output.

•In addition to being a source of wasted energy, leaks can also contribute to other operating losses. Leaks cause a drop in system pressure, which can make air tools function less efficiently, adversely affecting production. In addition, by forcing the equipment to run longer, leaks shorten the life of almost all system equipment (including the compressor package itself). Increased running time can also lead to additional maintenance requirements and increased unscheduled downtime. Finally, leaks can lead to adding unnecessary compressor capacity.

•While leakage can come from any part of the system, the most common problem areas are:

- •Couplings, hoses, tubes, and fittings
- •Pressure regulators
- •Open condensate traps and shut-off valves
- •Pipe joints, disconnects, and thread sealants.

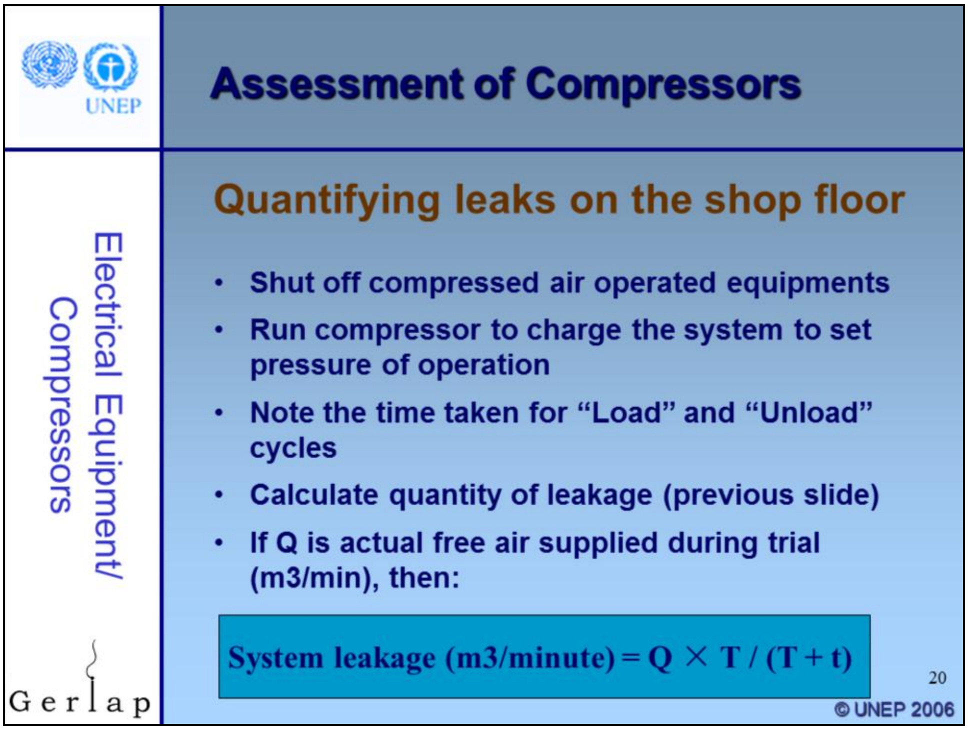
•Leakage rates are a function of the supply pressure in an uncontrolled system and increase with higher system pressures. Leakage rates identified in cubic feet per minute (cfm)



•For compressors that have start/stop or load/unload controls, there is an easy way to estimate the amount of leakage in the system. The method involves starting the compressor when there are no demands on the system. A number of measurements are taken to determine the average time it takes to load and unload the compressor.

•Total leakage in percentage can be calculated as: Leakage (%) = [(T x 199) / (T + t)], where T = on-load time, and t = off-load time.

•Leakage will be expressed in terms of the percentage of compressor capacity lost. The percentage lost to leakage should be less than 10per cent in a well maintained system. Poorly maintained systems can have losses as high as 20 to 30 percent of air capacity and power.



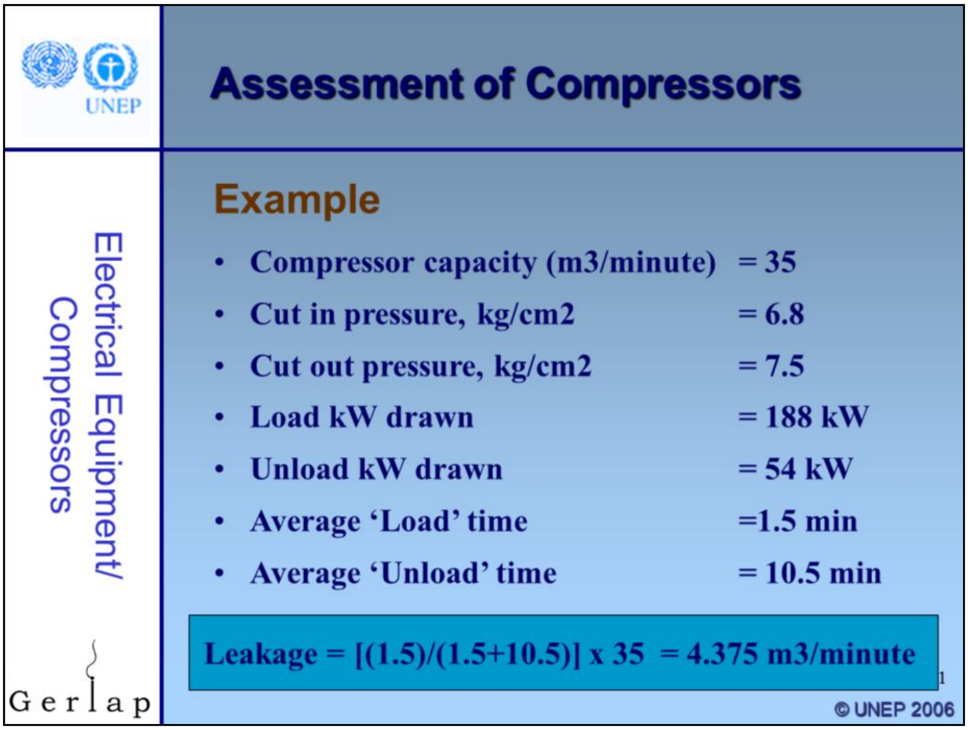
This is a simple method to quantify leaks in a compressed air system. These are the steps:

•Shut off compressed air operated equipments (or conduct a test when no equipment is using compressed air).

•Run the compressor to charge the system to set pressure of operation

•Note the subsequent time taken for "Load" and "Unload" Cycles of the compressors.

•Use the above expression to find out the quantity of leakage in the system. If Q is the actual free air being supplied during trial then the system leakage would be: System leakage = Q × T / (T + t)



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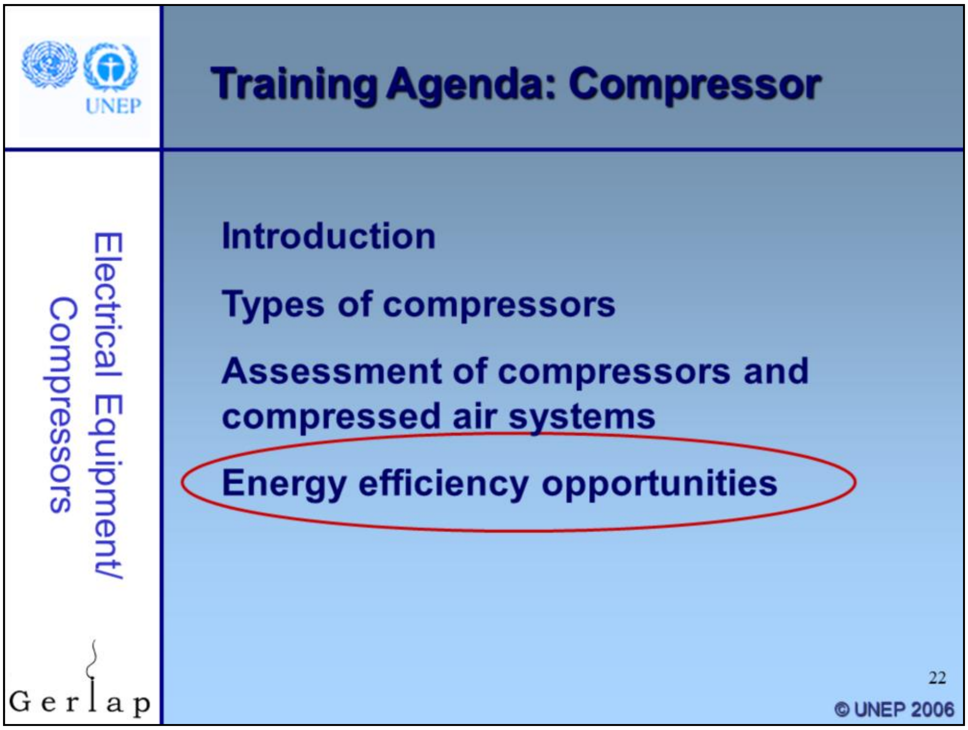
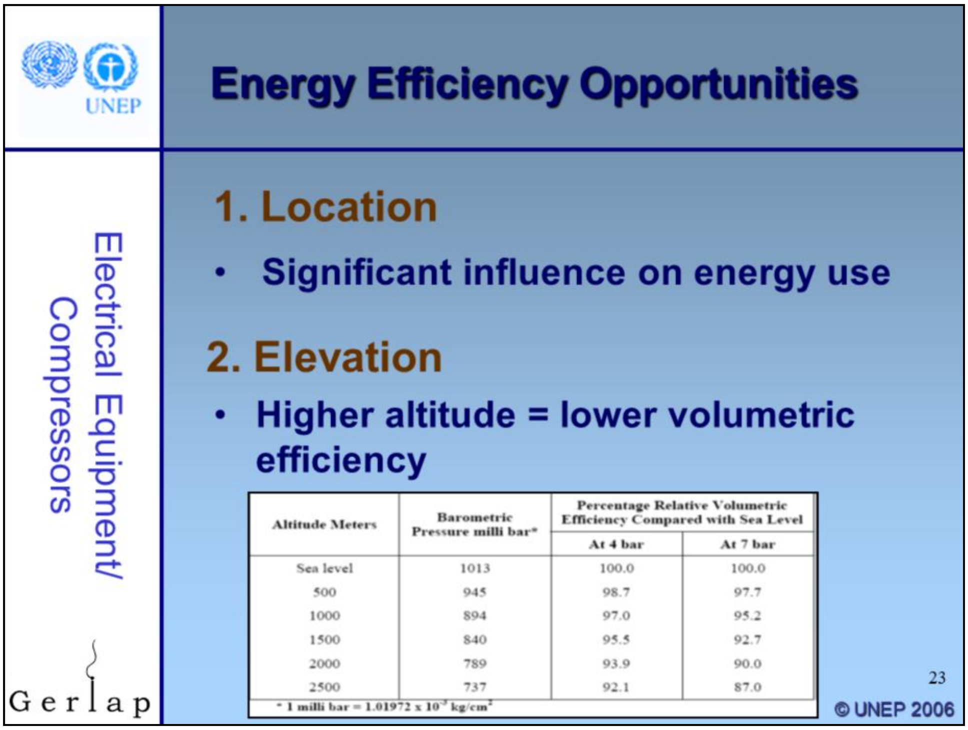


•Energy efficiency opportunities

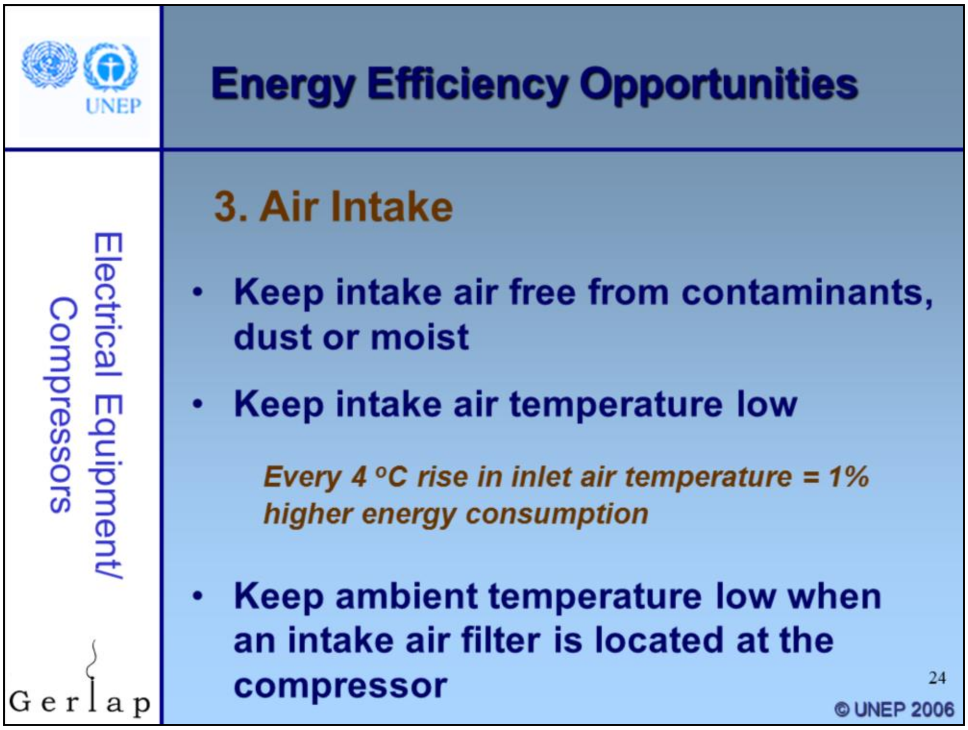


Altitude Meters	Barometric Pressure milli bar*	Percentage Relative Volumetric Efficiency Compared with Sea Level	
		At 4 bar	At 7 bar
Sea level	1013	100.0	100.0
500	945	98.7	97.7
1000	894	97.0	95.2
1500	840	95.5	92.7
2000	789	93.9	90.0
2500	737	92.1	87.0

* 1 milli bar = 1.01972 x 10^{-3} kg/cm^2

•The location of air compressors and the quality of air drawn by the compressors will have a significant influence on the amount of energy consumed. Compressor performance as a breathing machine improves with cool, clean, dry air at intake

•Altitude has a direct impact on the volumetric efficiency of a compressor. The effect of altitude on volumetric efficiency is given in the Table 6. It is evident that compressors located at higher altitudes consume more power to achieve a particular delivery pressure than those at sea level, as the compression ratio is higher.

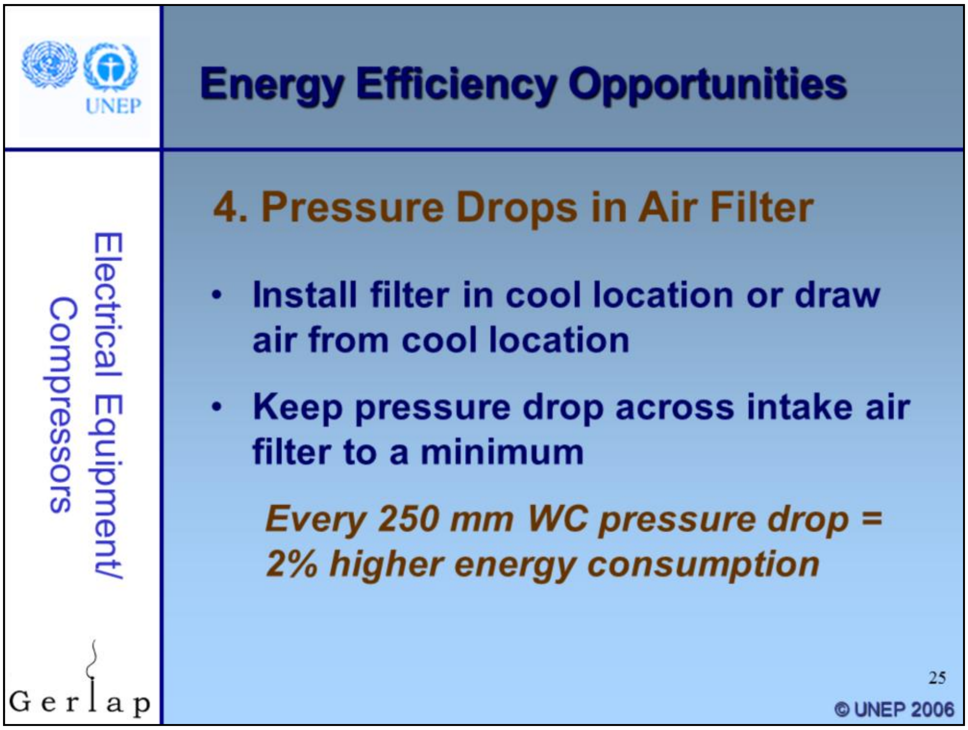


•The effect of intake air on compressor performance should not be underestimated.

•Intake air that is contaminated or hot can impair compressor performance and result in excess energy and maintenance costs. If moisture, dust, or other contaminants are present in the intake air. These contaminants can build up on the internal components of the compressor.

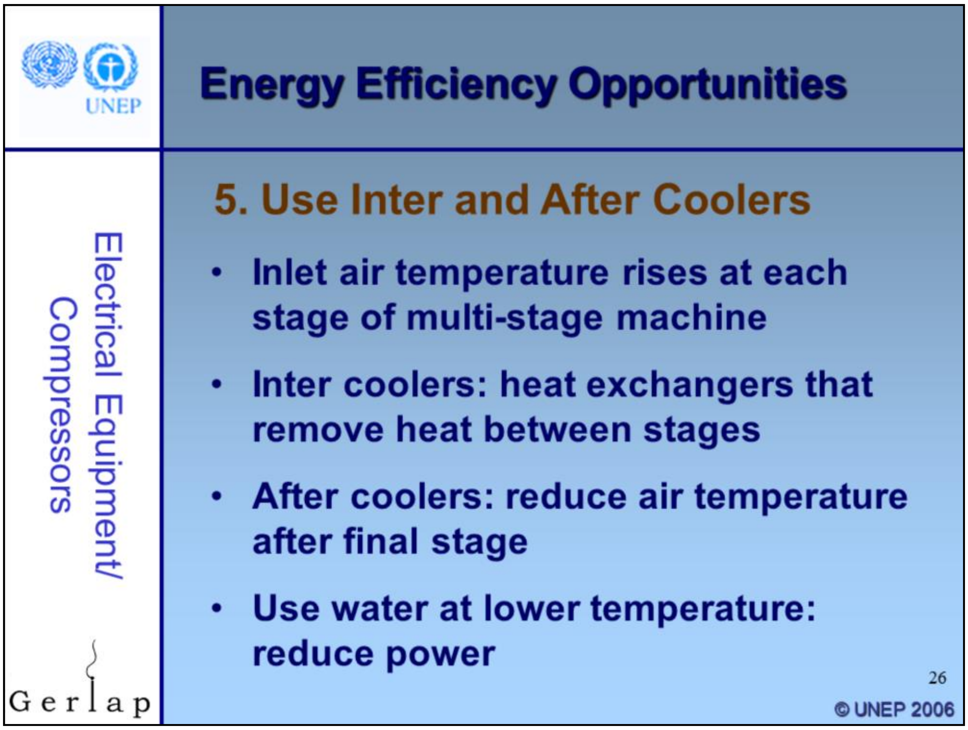
•The compressor generates heat due to its continuous operation. This heat gets dissipated to compressor chamber and leads to hot air intake. This results in lower volumetric efficiency and higher power consumption. As a general rule, *"Every 40C rise in inlet air temperature results in a higher energy consumption by 1per cent to achieve equivalent output"*. Hence cool air intake improves the energy efficiency of a compressor**.**

•When an intake air filter is located at the compressor, the ambient temperature should be kept to a minimum, to prevent reduction in mass flow. This can be accomplished by locating the inlet pipe outside the room or building. When the intake air filter is located outside the building, and particularly on a roof, ambient considerations may be taken into account.



•A compressor intake air filter should be installed in, or have air brought to it from a clean, cool location. The better the filtration at the compressor inlet, the lower the maintenance at the compressor.

•However, the pressure drop across the intake air filter should be kept to a minimum. The pressure drop across a new inlet filter should not exceed 3 pounds per square inch. As a general rule "*For every 250 mm WC pressure drop increase across at the suction path due to choked filters etc, the compressor power consumption increases by about 2per cent for the same output*" .

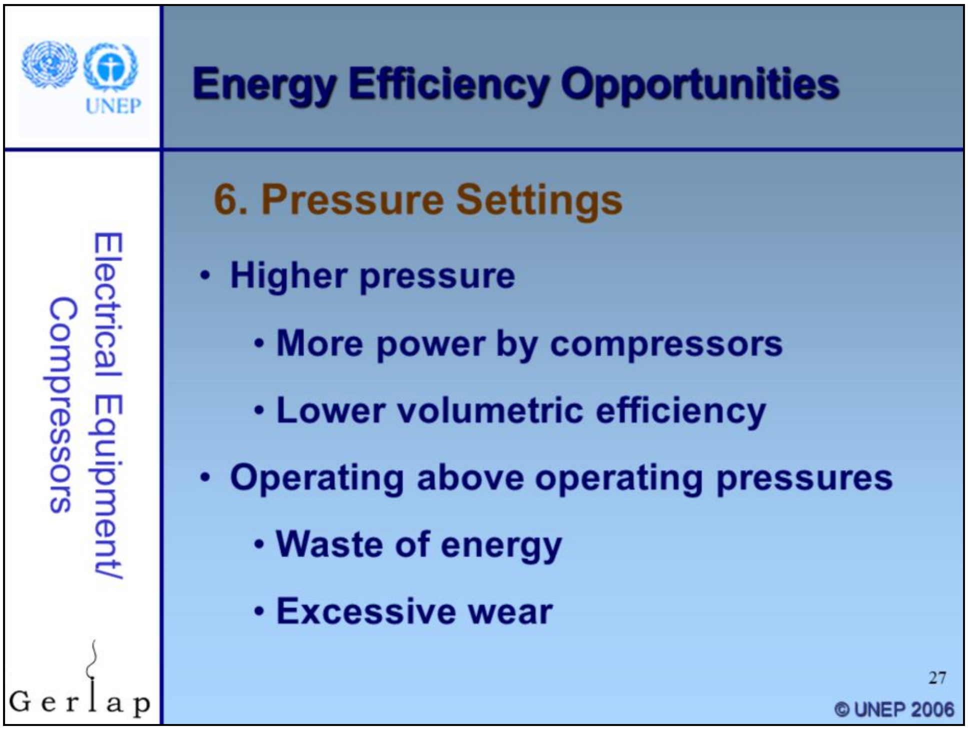


•**Perfect cooling.** Ideally, the temperature of the inlet air at each stage of a multi-stage machine should be the same as it was at the first stage. This is referred to as "perfect cooling" or isothermal compression. But in actual practice, the inlet air temperatures at subsequent stages are higher than the normal levels.

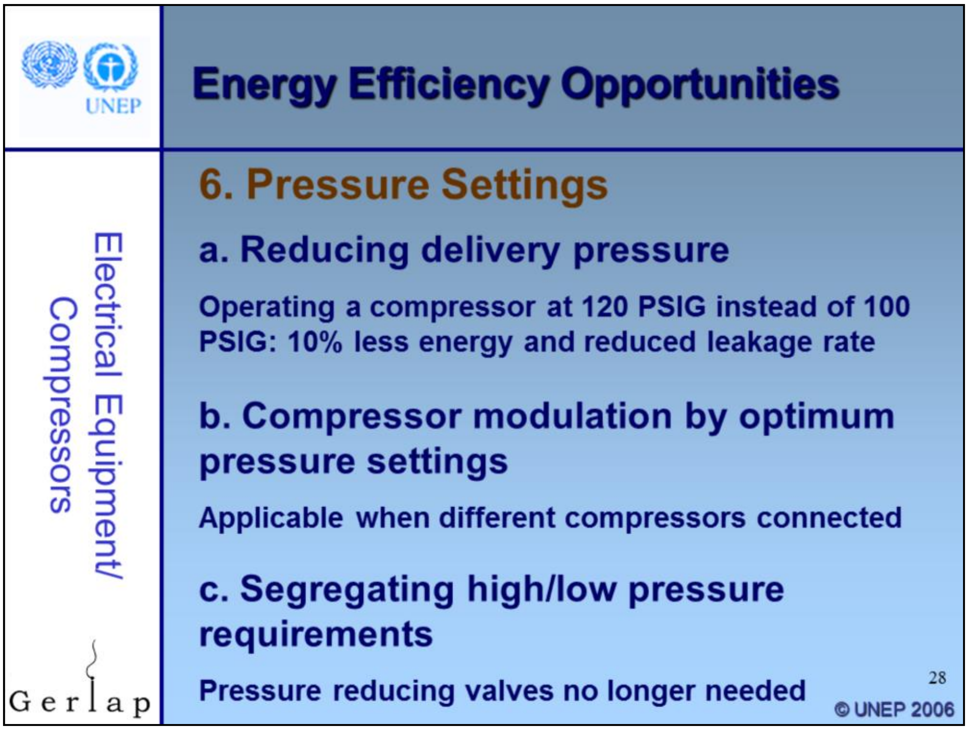
•**Intercoolers.** Most multi-stage compressors use intercoolers. These are heat exchangers that remove the heat of compression between the stages of compression. Intercooling affects the overall efficiency of the machine.

•**After-coolers.** As mechanical energy is applied to a gas for compression, the temperature of the gas increases. After-coolers are installed after the final stage of compression to reduce the air temperature. As the air temperature is reduced, water vapor in the air is condensed, separated, collected, and drained from the system.

•Use of water at lower temperature reduces specific power consumption. However, very low cooling water temperature could result in condensation of moisture in the air, which if not removed would lead to cylinder damage.



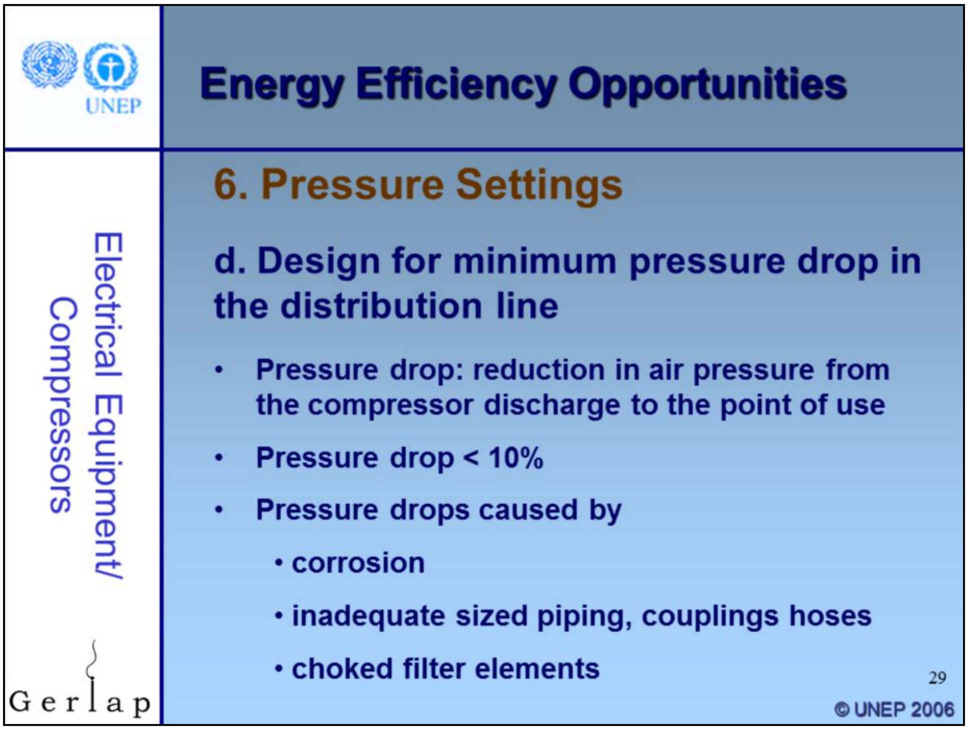
•For the same capacity, a compressor consumes more power at higher pressures. Subsequently, compressors should not be operated above their optimum operating pressures as this not only wastes energy, but also leads to excessive wear, leading to further energy wastage. The volumetric efficiency of a compressor is also less at higher delivery pressures.



•**Reducing delivery pressure.** The possibility of lowering and optimizing the delivery pressure settings should be explored by a careful study of pressure requirements. The operating of a compressed air system gently affects the cost of compressed air. Operating a compressor at 120 PSIG instead of 100 PSIG, for instance, requires 10 per cent more energy as well as increasing the leakage rate. Therefore, every effort should be made to reduce the system and compressor pressure to the lowest possible setting.

•***(click once)*** **Compressor modulation by optimum pressure settings.** Very often in an industry, different types, capacities and makes of compressors are connected to a common distribution network. In such situations, proper selection of a right combination of compressors and optimal modulation of different compressors can conserve energy. For example, where more than one compressor feeds a common header, compressors have to be operated in such a way that the cost of compressed air generation is minimal.

•***(click once)*** **Segregating high and low pressure requirements.** If the low-pressure air requirement is considerable, it is advisable to generate low pressure and high-pressure air separately and feed to the respective sections instead of reducing the pressure through pressure reducing valves, which invariably waste energy.



•Pressure drop is a term used to characterize the reduction in air pressure from the compressor discharge to the actual point-of-use. Pressure drop occurs as the compressed air travels through the treatment and distribution system.

•A properly designed system should have a pressure loss of much less than 10per cent of the compressor's discharge pressure, measured from the receiver tank output to the point-of-use. The longer and smaller diameter the pipe is, the higher the friction loss.

•Pressure drops are caused by corrosion and the system components themselves are important issues. Excess pressure drop due to inadequate pipe sizing, choked filter elements, improperly sized couplings and hoses represent energy wastage.

Energy Efficiency Opportunities

6. Pressure Settings

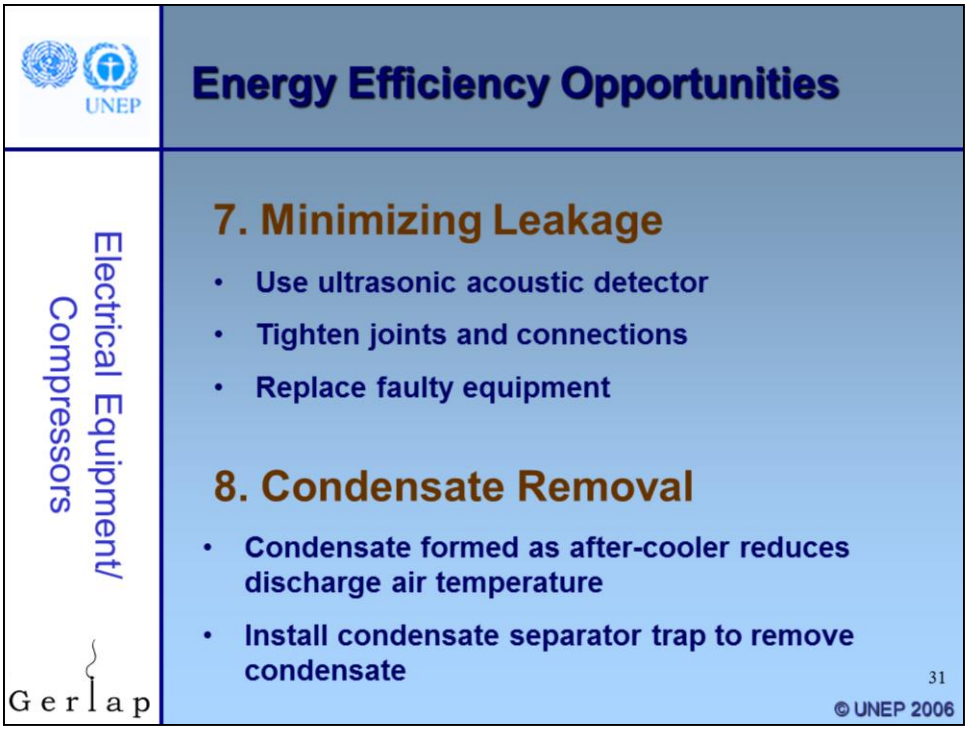
d. Design for minimum pressure drop in the distribution line

Pipe Nominal Bore (mm)	Pressure drop (bar) per 100 meters	Equivalent power losses (kW)
40	1.80	9.5
50	0.65	3.4
65	0.22	1.2
80	0.04	0.2
100	0.02	0.1

Typical pressure drop in compressed air line for different pipe size (Confederation of Indian Industries) 30

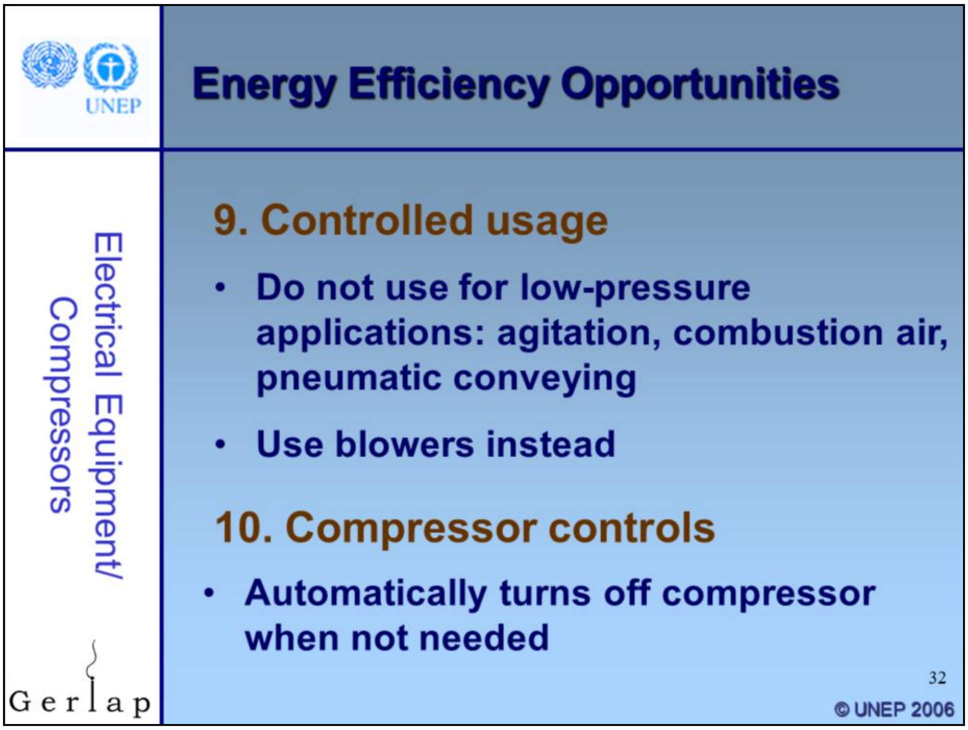
This table illustrates the energy wastage, if the pipes are of smaller diameter.

Typical acceptable pressure drop in industrial practice is 0.3 bar in the mains header at the farthest point and 0.5 bar in distribution system.



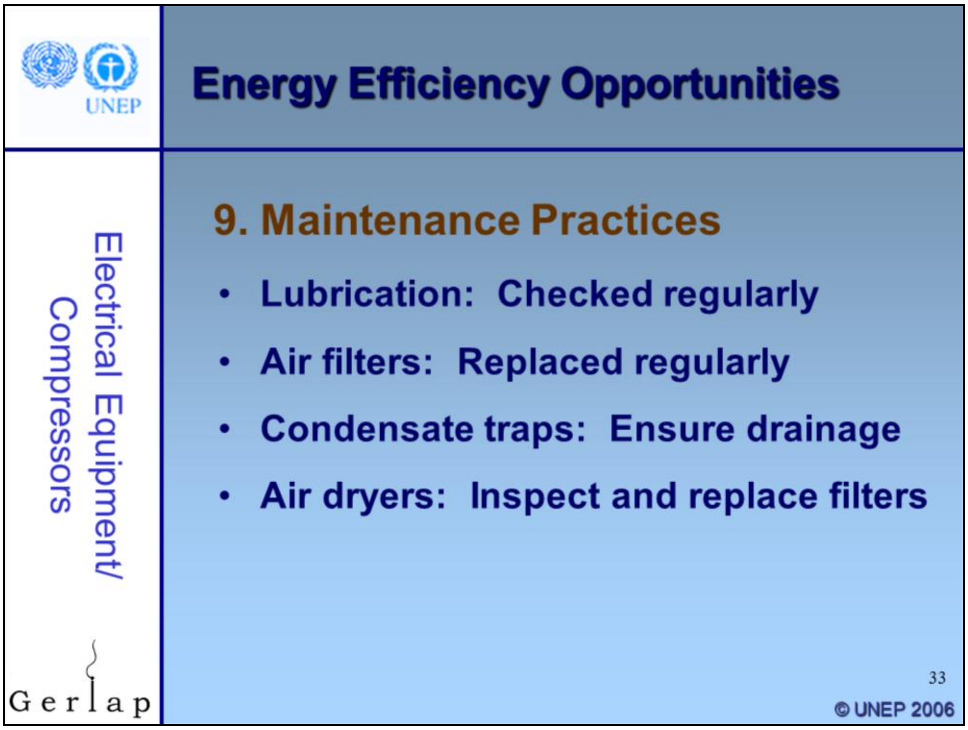
•**Minimizing leakage.** Compressed air leakage accounts for substantial power wastage. The best way to detect leaks is to use an ultrasonic acoustic detector that can recognize the high-frequency hissing sounds associated with air leaks. Leaks occur most often at joints and connections. Stopping leaks can be as simple as tightening a connection or as complex as replacing faulty equipment.

•***(click once)*** **Condensate removal.** After compressed air leaves the compression chamber the compressor's after-cooler reduces the discharge air temperature well below the dew point. Therefore, considerable water vapor is condensed. To remove this condensation, most compressors with built-in after-coolers are furnished with a combination condensate separator-trap.



•**Controlled usage.** Since the compressed air system is already available, plant engineers may be tempted to use compressed air to provide air for low-pressure applications such as agitation, pneumatic conveying or combustion air. Using a blower that is designed for lower pressure operation will cost only a fraction of compressed air generation energy and cost. *Note: in some companies, staff use compressed air to clean their clothes. Apart from the energy wastage, this is a very dangerous practice.*

•**Compressor controls.** Air compressors become inefficient when they are operated at significantly below their rated cfm output. To avoid running extra air compressors when they are not needed, a controller can be installed to automatically turn compressors on and off, based on demand. Also, if the pressure of the compressed air system is kept as low as possible, efficiency improves and air leaks are reduced.



•Good and proper maintenance practices will dramatically improve the performance efficiency of a compressor system. Here are a few tips for efficient operation and maintenance of industrial compressed air systems:

•Lubrication: Compressor oil pressure should be visually checked daily, and the oil filter changed monthly.

•Air Filters: The inlet air filter can easily become clogged, particularly in dusty environments. Filters should be checked and replaced regularly.

•Condensate Traps: Many systems have condensate traps to gather flush condensate from the system. Manual traps should be periodically opened and re-closed to drain any accumulated fluid and automatic traps should be checked to verify they are not leaking compressed air.

•Air Dryers: Drying air is energy-intensive. For refrigerated dryers, inspect and replace pre-filters regularly as these dryers often have small internal passages that can become plugged with contaminants.

Training Session on Energy Equipment

Electrical Equipment/ Compressors

Compressors & Compressed Air Systems

THANK YOU FOR YOUR ATTENTION

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